

Matrix Inverse via LU factorization

input : $n \times n$ matrix $[A]$

output: $[A]^{-1}$

$[L], [U] \leftarrow \text{lu_decompose}([A])$

for $k \leftarrow 1$ **to** n **do**

$\{e_k\} \leftarrow$ unit vector with 1 at element k ; 0 elsewhere
 $A^{-1}[:, k] \leftarrow \text{lu_solve}([L], [U], \{e_k\})$

end